GAURAV SINGH

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EDUCATION

Indian Institute of Technology Jodhpur | B.Tech, Electrical Engineering | **CGPA: 7.66 / 10** 2024 – 2028
New Horizon Public School, Panvel | Class XII (CBSE) | 95.4% | School Rank 1 2023

ACHIEVEMENTS

- **LeetCode Knight** with a contest rating of **1850+**.
- Solved **750+ DSA problems** across LeetCode, GeeksforGeeks, and Codeforces.

PROJECTS

Mini Redis: Redis-Inspired In-Memory Database github.com/gaurav-singh2525/redis-inspired-db
C++17, TCP Sockets, Multithreading, File I/O

- Built a C++17 in-memory key-value store reaching **95K ops/sec** under concurrent load (35K GET ops/sec single-client), using an 8-thread producer-consumer pool (mutex + condition variables) and a custom RESP parser (SET, GET, DEL, EXPIRE, TTL, COUNT).
- Added durability with snapshots and **Write-Ahead Logging**, **recovering 100K keys in 170 ms** through log replay and checkpoint restoration.
- Implemented active + lazy TTL expiration and **O(1) LRU eviction** (hash map + doubly linked list), sustaining **50K+ insertions** under enforced memory limits; profiled throughput, latency, and crash recovery with config-driven benchmarks.

AsyncTasker: Fault-Tolerant Distributed Job Queue github.com/gaurav-singh2525/AsyncTasker
Node.js, Express.js, Redis, PostgreSQL

- Developed a **distributed job queue** on Redis priority queues with PostgreSQL persistence, guaranteeing at-least-once execution at **1,093 jobs/sec** (10 workers) with **sub-10 ms P99** API latency.
- Automated **crash recovery** that rebuilds queue state from persistent storage and reclaims abandoned in-flight jobs, restoring **10K jobs at 2,176+ jobs/sec**.
- Built a concurrent worker pool with configurable parallelism, timeouts, and graceful job lifecycle handling; added priority-aware delayed scheduling and exponential-backoff retries on Redis Sorted Sets.

Cloud-Native LLM API Gateway github.com/gaurav-singh2525/Cloud-Native-API-gateway
Go, Redis Stack, OpenAI API, Docker, Kubernetes, GitHub Actions

- Built a **high-concurrency LLM API gateway in Go** with dynamic reverse proxying, request context cancellation, and fail-fast Bearer-token authentication middleware.
- Cut upstream LLM cost and latency with a **semantic cache**: OpenAI embeddings + Redis Stack KNN search (cosine distance) skip LLM calls for semantically equivalent prompts.
- Enforced per-minute token budgets with **distributed, token-aware rate limiting** using atomic Redis pipelines (INCRBY + EXPIRE), computed from prompt payload size.
- Productionized with a **15 MB** multi-stage Docker image, Kubernetes liveness/readiness probes, and GitHub Actions CI/CD for zero-downtime rolling deployments.

Real-Time Collaborative Code Platform github.com/gaurav-singh2525/collab_platfrom
Go (Gin), React, PostgreSQL, Docker, WebSockets

- Developed a **full-stack collaborative coding platform** with live code sync, presence tracking, and shared execution output over low-latency WebSockets, using a Monaco Editor + React frontend and Go backend.
- **Sandboxed multi-language execution** (Python, JavaScript, C++) in Docker containers with enforced CPU, memory, network, and time limits.
- Designed a room-based session model with PostgreSQL persistence and **JWT-authenticated REST APIs** for workspace creation, rejoining, and state continuity.

TECHNICAL SKILLS

Languages: C++, Go, Python, JavaScript, C
Backend & Systems: Go (Gin, net/http), Node.js, Express.js, REST APIs, WebSockets, Concurrency, Multithreading, TCP/IP & Socket Programming
Databases: PostgreSQL, Redis
Cloud & DevOps: Docker, Kubernetes, CI/CD (GitHub Actions)
Core CS: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, DBMS
Frontend & Tools: React, Git, Linux, Postman

LEADERSHIP & EXTRACURRICULARS

- **Core Member, Programming Club, IIT Jodhpur:** co-organized competitive programming contests and workshops.
- **Student Guide, Student Well-being Committee:** mentored 14 freshers academically and non-academically.
- **Support Team, Career Development Cell (CDC):** assisted Placement Co-coordinators in running placement processes.